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%% Retrieiving raw data from .txt file saved from pico-scope software
folder = uigetdir('****'); %Select folder file is in
cd(folder);
filename = uigetfile('****',folder); %Select .txt file
data = dlmread(filename, ' ', 2, 0); % Read raw data into matrix format
data(:, 1) = data(:, 1) * 1000; %convert milliseconds to microseconds
color = [];
color = jet; %Select a colormap from which to choose color for plot curves/markers.
col = color(1, :); %Select the particular color from the Matlab colormap specified above.

%% Physical properties of the bar and the sample
E_s = 93 * 10^9; %sample Young's modulus (Pa)
E_b = 186.5328 * 10^9; %Bar Young's Modulus (Pa)
l = 0.0291; %sample axial length
D = 0.0291; %sample diameter (m)
p_s = 2817; %sample density (kg/m^3)
p_b = 8056; %Bar density (kg/m^3)

C_s = sqrt(E_s / p_s); %P-wave speed in sample (m/s)
C_b = sqrt(E_b / p_b); %P-wave speed in bar (m/s)
t_delay = (10^6) * l / C_s; %travel time delay between reflected and transmitted pulses due to travel through sample (microseconds)
A_b = pi * ((38.1 * 10^-3) / 2)^2; %Area of bar (m^2)
A_s = pi * ((D) / 2)^2; %Area of sample (m^2)
igdist = 1.27635; %Distance from incident bar/sample interface to inc gauge (m)
tgdist = 1.27318; %Distance from trans bar/sample interface to trans gauge (m)

%% Resetting voltage signals to true zero
data(:, 2) = data(:, 2) - mean(data(500:2500, 2)); %corrects from offset in level of background
data(:, 3) = data(:, 3) - mean(data(500:2500, 3)); %corrects from offset in level of background
iV = data(:, 2); tV = data(:, 3); %Incident (iV) and transmitted (tV) voltage (V)
t = data(:, 1); %time (t) (microseconds)

%% Smoothing voltage data due to poor resolution of 3 strain gauge receiving of equipment
rV = smooth(rV, 2000);
iV = smooth(iV, 2000);
tV = smooth(tV, 2000);

%% Raw Data Axial Plot Figure 1
figure(1); clf('reset'); hold on;
plot(t, iV, 'LineWidth', 1, 'Color', 'blue'); %plot incident bar voltage time series
plot(t, tV, 'LineWidth', 1, 'Color', 'red'); %plot transmitted bar voltage time series
grid on
xlabel('Time (microseconds)'); ylabel('Voltage, V'); legend('Incident', 'Transmitted');

%Visual selection of incident pulse start and end times
txt1 = annotation('textbox', 'string', 'Zoom in to incident pulse. Press any key to exit zoom and select time');
zoom on; pause; delete(txt1);
txt2 = annotation('textbox', 'string', 'Select start and end times of incident pulse');
[X1, Y1] = ginput(2); zoom off; zoom out; delete(txt2); %select start and end times of incident pulse by clicking on
appropriate parts of raw incident and transmitted bar time series

%% Determining starting time of each pulse
tStep = mean(diff(t)); %8 * 10^-3; % Time Step (microseconds)
t_I = X1(1); %start time of incident signal (microseconds)

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t_R = t_I + (2 * igdist / C_b) * 10^6; %start time of reflected signal (microseconds)
t_T = t_I + (igdist / C_b + l / C_s + tgdist / C_b) * 10^6; %start time of transmitted signal (microseconds)
dur = (X1(2) - X1(1)); %pulse duration (microseconds)

%% incident pulse Figure 2
%indexing selected data by time intervals
[c(1) ind(1)] = min(abs(t_I - t)); %index of start time of incident pulse
[c(2) ind(2)] = min(abs(X1(2) - t)); %index of end time of incident pulse
[c(3) ind(3)] = min(abs(t_R - t)); %index of start time of reflected pulse
[c(4) ind(4)] = min(abs(t_T - t)); %index of start time of transmitted pulse

index = (ind(1):ind(2)); %grab all indices between start and end of incident pulse
time = t(ind(1)) - t_I; %zero time associated with indexed times to start time of incident pulse
inc = iV(index); %pull out voltages associated with indexed times from incident pulse duration
trans = tV(index + (ind(4) - ind(1))); %transmitted data at indices = (times from incident pulse duration + transmitted
time - start of incident pulse time)
ref = iV(index + (ind(3) - ind(1)));

figure(2)
clf(2)
plot(time, smooth(inc, 500), 'LineWidth', 3, 'Color', 'blue'); xlabel('Time (ms)'); ylabel(' voltage, V'); hold on;
%% Transmitted pulse Figure 2
plot(time, smooth(trans, 500), 'LineWidth', 3, 'Color', 'red');

%% Reflected pulse Figure 2
plot(time, smooth(ref, 500), 'LineWidth', 1.5, 'Color', 'blue');
plot(time, smooth(inc + ref, 500), 'LineWidth', 3, 'Color', 'black');
legend('Incident', 'Transmitted', 'Reflected', 'Incident + Reflected');

diseq = 100 * abs(smooth(inc) - smooth(ref) - smooth(trans)) / mean(trans); %instantaneous percent difference between
the forces at the two specimen faces
[X, Y] = ginput(2)
meandiff = mean(diseq((round(X(1)/(tStep)):round(X(2)/(tStep))))); %mean value of the percent difference in force
at specimen faces.
meandiff = ['Mean difference = ', num2str(meandiff), ' %']; %Energy of the incident pulse
%annotation('textbox', 'string', meandiff)

%% converting volt to strain from formula
sgR = 120; % Strain Gauge Resistance
Vcal = 2; % Gauge calibration voltage
Rcal = 59940; % Gauge calibration resistance
GF = 2.14; % Gauge factor
trans = (trans * sgR) / (Vcal * GF * (sgR + Rcal)); %converting transmitted pulse to bar strain from Volt
inc = (inc * sgR) / (Vcal * GF * (sgR + Rcal)); %converting incident pulse to bar strain from Volt
ref = (ref * sgR) / (Vcal * GF * (sgR + Rcal)); %converting reflected pulse to bar strain from Volt

%Sample Axial Strain Rate
strain_rate = [];
strain_rate(:, 1) = 2 * (C_b / l) * (ref); %one wave method to determine strain rate
%strain_rate(:, 1) = (C_b / l) * (inc - ref - trans); %three wave method to determine strain rate

%index1 = find(strain_rate < 0);
% strain_rate(index1) = 0;

strain = [];
delt = tStep * 10^-6; %Time Step (s)

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for i=1:length(strain_rate(:,1))
    strain(i)= sum(strain_rate(1:i)).*delt;
end

sig=[];
sig=(A_b/A_s)*E_b*(trans); %Pa

%close all
%strain vs time
figure(3)
subplot(3,1,1),plot(time,smooth(strain,500),'LineWidth',3,'color',col); grid on;
xlabel('Time (\mus)'); ylabel('\epsilon_a Axial Strain'); %ylim([-0.005 0.00005]); xlim([min(time+.02),
max(time/2.25)]);
hold on;

%axial strain rate vs time
subplot(3,1,2), plot(time, smooth(strain_rate,500),'LineWidth',3,'color',col); grid on;
xlabel('Time (\mus)'); ylabel('\delta\epsilon_a Axial Strain Rate'); %ylim([-250 5]); xlim([min(time+.02),
max(time/2.25)]);
hold on;

%axial stress vs time
subplot(3,1,3),plot(time,smooth(-sig*10^-6,500),'LineWidth',3,'color',col);grid on;
xlabel('Time (\mus)');ylabel(' \sigma_a Axial Stress (MPa)'); %ylim([-80 5]); xlim([min(time+.02),
max(time/2.25)]);
hold on;

%axial stress vs axial strain
figure(4)
plot(strain, -sig*10^-6, xlabel('\epsilon'), ylabel('\sigma (MPa)')

%%%Calculating toughness, U_t (J/m^3) through integration of stress-strain curve
U_t = trapz(strain,sig)*10^6 %toughness (J/m^3)

%%%Total energy absorbed in specimen (J)
Etot = U_t*(pi*(D/2)^2*L)
Etot = ['Etot = ', num2str(Etot), ' J']; %Convert to string
annotation('textbox','string',Etot) %Annotate on stress-strain plot

%Power traveling through the bar for each pulse at time, t. (watts)
P_I= (((0.5*p_b*A_b*C_b^3)+(0.5*A_b*C_b*E_b)).*((-1*inc).^2)
P_R= (((0.5*p_b*A_b*C_b^3)+(0.5*A_b*C_b*E_b)).*((-1*ref).^2)
P_T= (((0.5*p_b*A_b*C_b^3)+(0.5*A_b*C_b*E_b)).*((-1*trans).^2)
%Finding the energy of a pulse: integrate power over the whole pulse
%(joules)
E_i = trapz(time*10^-6,P_I)
E_r = trapz(time*10^-6,P_R)
E_t = trapz(time*10^-6,P_T)
E_d = E_i - E_r - E_t

E_I = ['E_I = ', num2str(E_i), ' J']; %Energy of the incident pulse
E_R = ['E_R = ', num2str(E_r), ' J']; %Energy of the reflected pulse
E_T = ['E_T = ', num2str(E_t), ' J']; %Energy of the transmitted pulse
E_d = ['E_d = ', num2str(E_d), ' J']; %Energy consumed in sample (kinetic, heat, fracture, plasticity, etc.)

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figure(2)
annotation('textbox','string',E_I)
annotation('textbox','string',E_R)
annotation('textbox','string',E_T)
annotation('textbox','string',E_d)
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